

Cylinder Head — LH

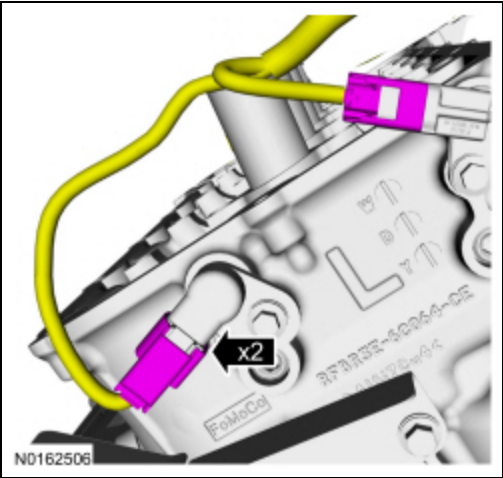


Material	
Item	Specification
Motorcraft® SAE 5W-20 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-20 Super Premium Motor Oil (Canada) XO-5W20-QSP (US); CXO-5W20-LSP12 (Canada)	WSS-M2C945-A

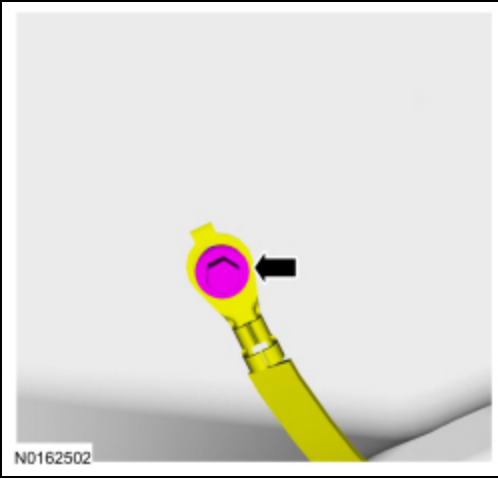
Removal

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces, that enters the oil passages, coolant passages or the oil pan, can cause engine failure.

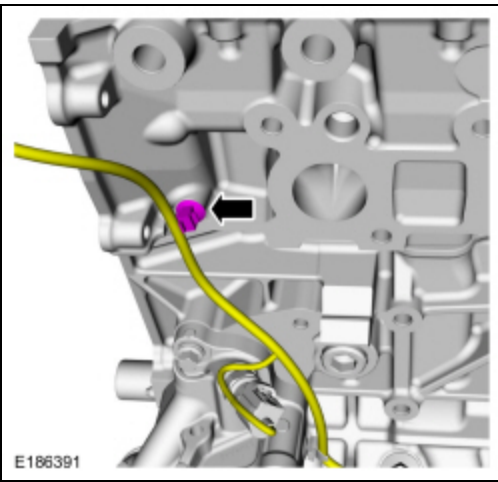
1. With the vehicle in NEUTRAL, position it on a hoist. Refer to Section 100-02.
2. Release the fuel system pressure. Refer to Section 310-00.
3. Drain the engine cooling system. Refer to Section 303-03.
4. Remove the ~~ACL~~. Refer to Section 303-12.
5. Remove the intake manifold. Refer to Intake Manifold.
6. Remove the ~~LH~~ hydraulic lash adjusters. Refer to Hydraulic Lash Adjuster.
7. Remove the ~~LH~~ exhaust manifold. Refer to Section 309-00.
8. Disconnect the ~~LH CMP~~ sensor electrical connectors.



9. Remove the bolt from the cowl and position the ground strap aside.



10. Detach the wiring harness retainer.



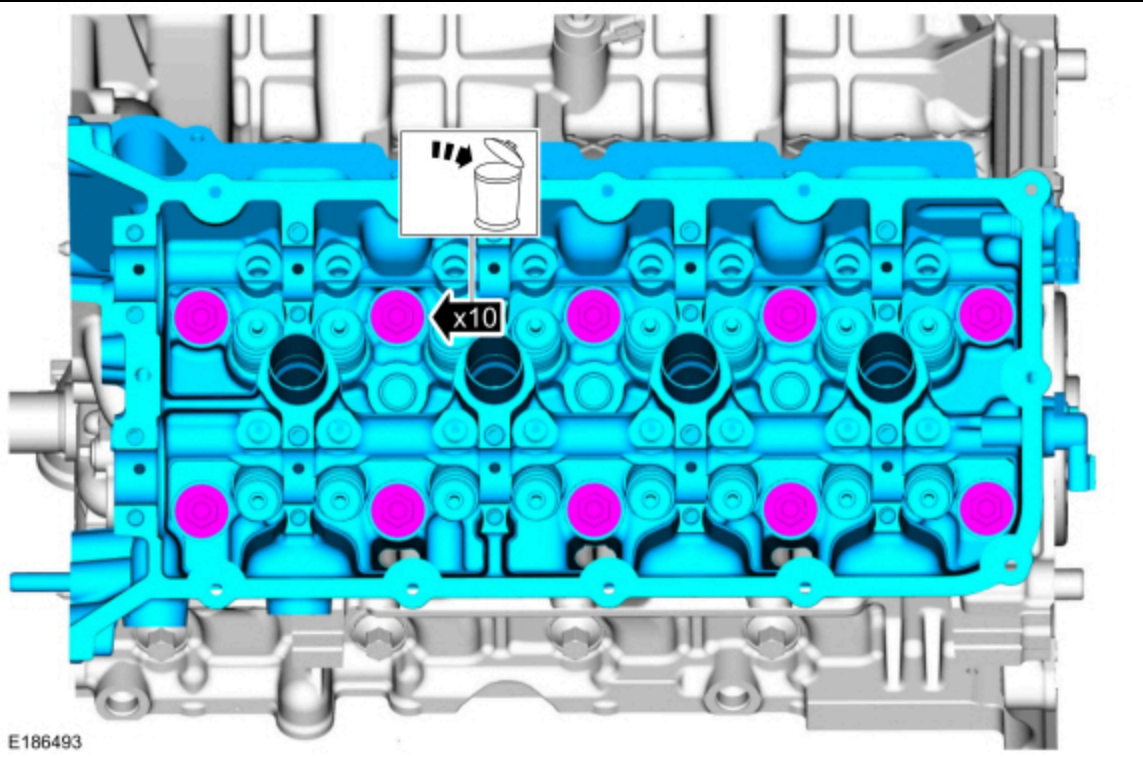
11. **NOTICE:** The cylinder head must be cool before removing it from the engine. Cylinder head warpage can result if a warm or hot cylinder head is removed.

NOTICE: Place clean shop towels over exposed engine cavities. Carefully remove the towels so foreign material is not dropped into the engine.

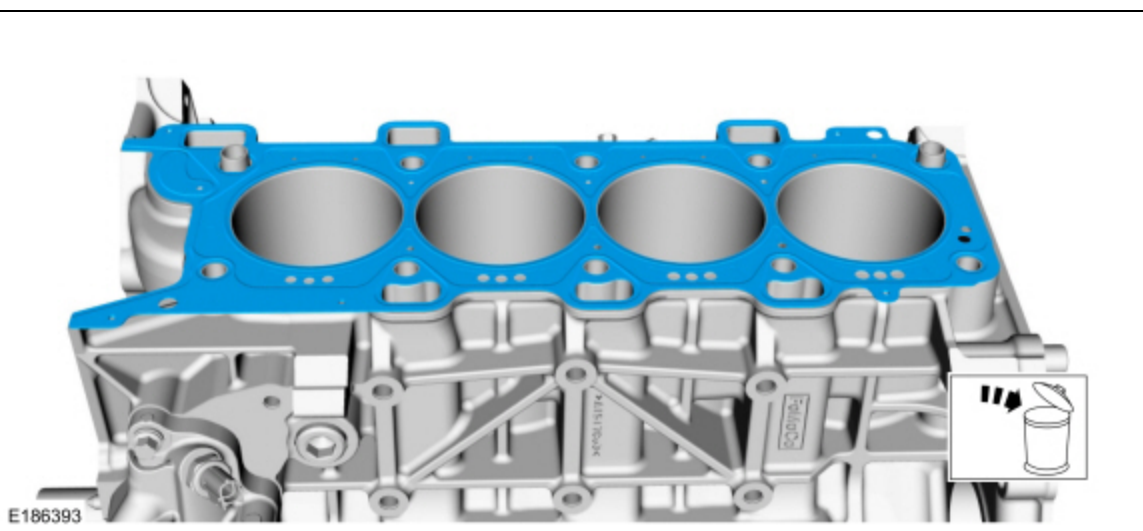
NOTICE: The cylinder head bolts must be discarded and new bolts must be installed. They are a tighten-to-yield design and cannot be reused.

Remove and discard the cylinder head bolts.

- Remove the cylinder head.



12. Remove and discard the cylinder head gasket.

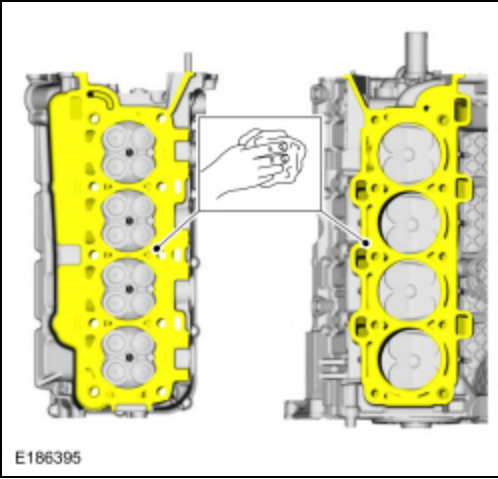


13. **NOTICE:** Do not use metal scrapers, wire brushes, power abrasive discs or other abrasive means to clean the sealing surfaces. These tools cause scratches and gouges which make leak paths.

NOTE: Clean the cylinder head bolt holes in the cylinder block using compressed air. Make sure all coolant, oil or other foreign material is removed.

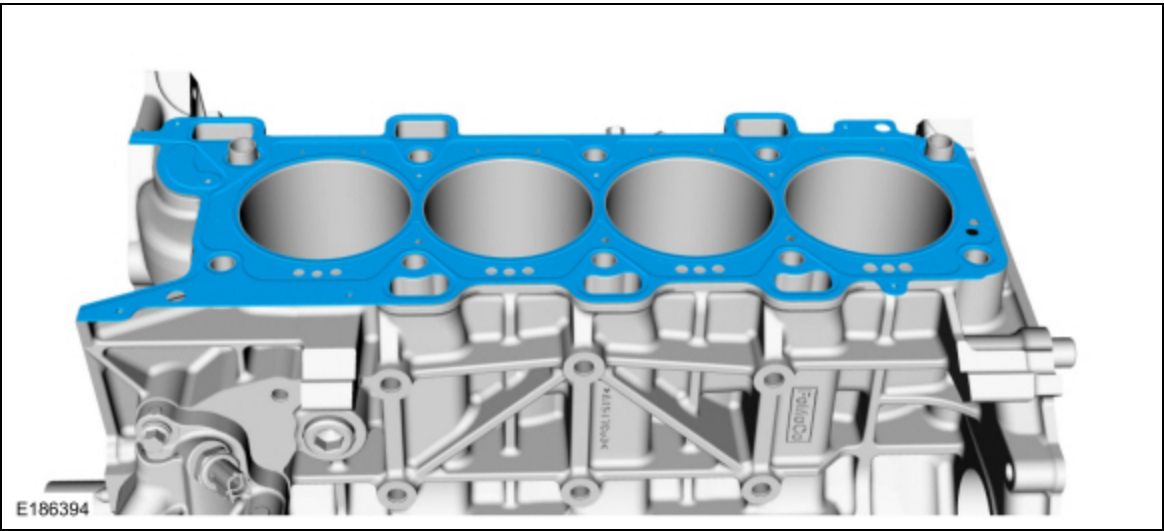
Clean and inspect the cylinder head-to-cylinder block mating surfaces of both the cylinder head and the cylinder block.

- Check the cylinder head distortion and the cylinder block distortion, paying particular attention to the oil pressure feed area. Refer to Section 303-00.



Installation

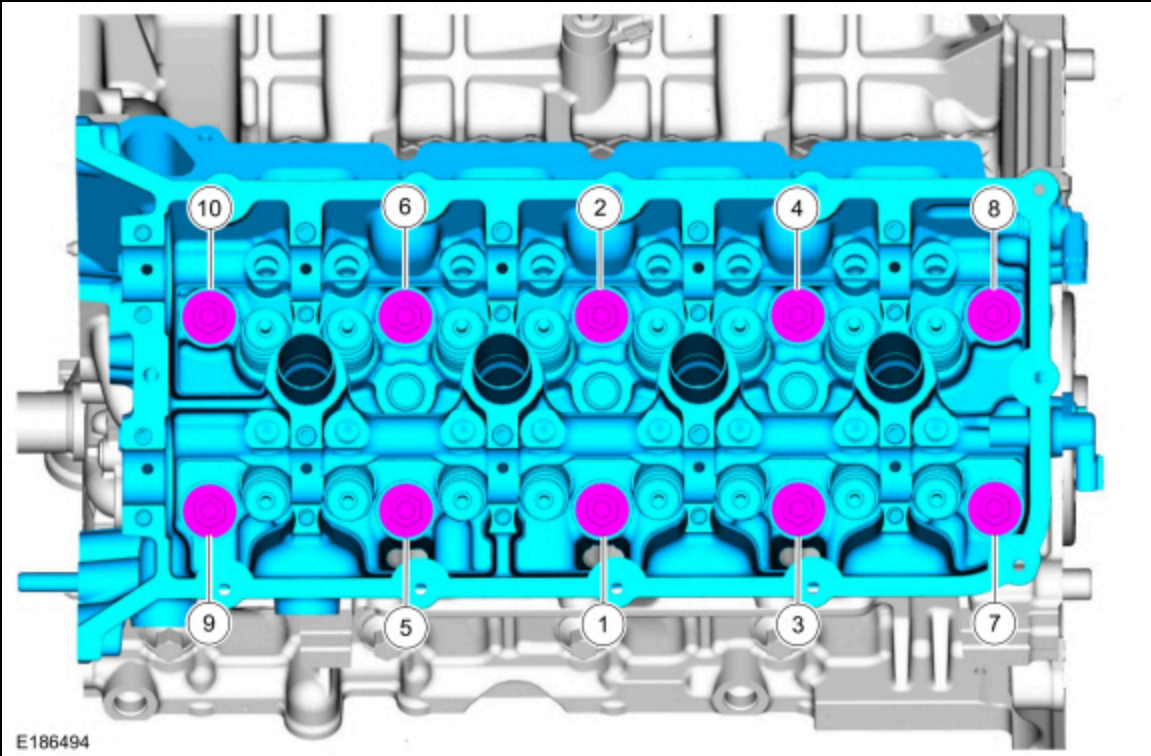
1. Install a new cylinder head gasket.



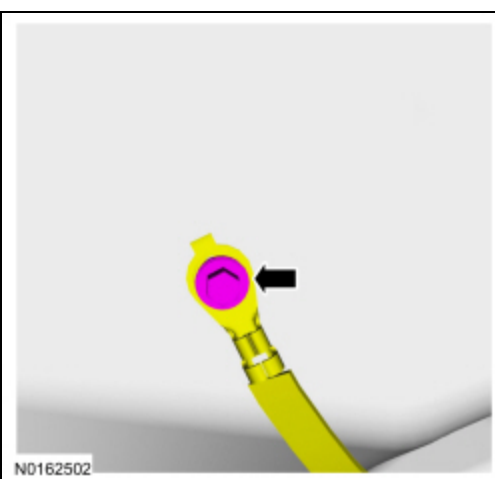
2. **NOTE:** *The cylinder head bolts are torque-to-yield and must not be reused. New cylinder head bolts must be installed.*

Install the cylinder head and the new bolts.

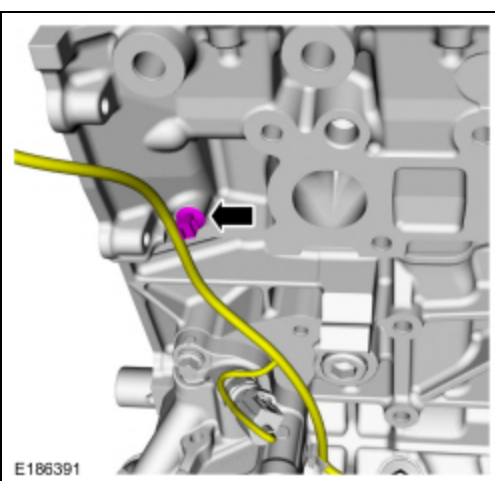
- Stage 1: Tighten to 25 Nm (18 lb-ft).
- Stage 2: Tighten to 75 Nm (55 lb-ft).
- Stage 3: **Loosen** 180 degrees.
- Stage 4: Tighten to 25 Nm (18 lb-ft).
- Stage 5: Tighten to 40 Nm (30 lb-ft).
- Stage 6: Tighten an additional 90 degrees.
- Stage 7: Tighten an additional 90 degrees.



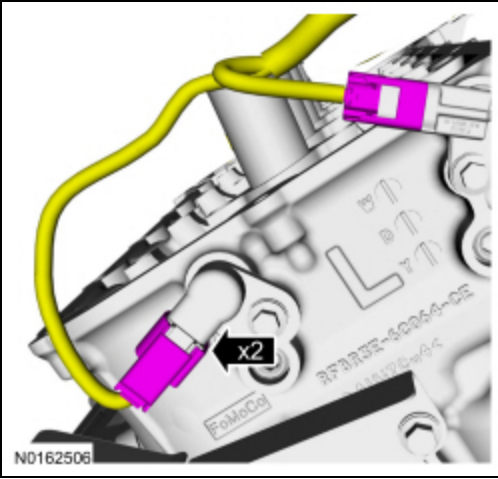
3. Install the ground strap and bolt onto the cowl.
 - Tighten to 10 Nm (89 lb-in).



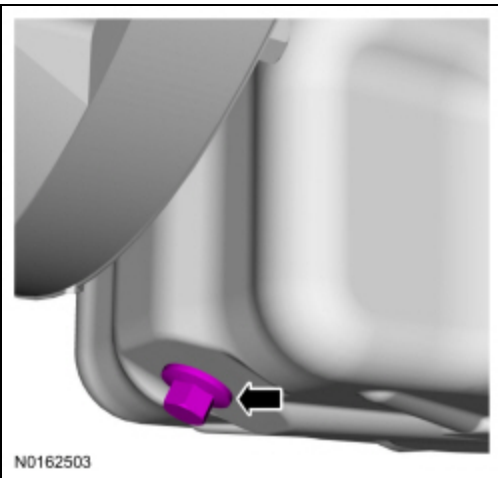
4. Attach the wiring harness retainer.



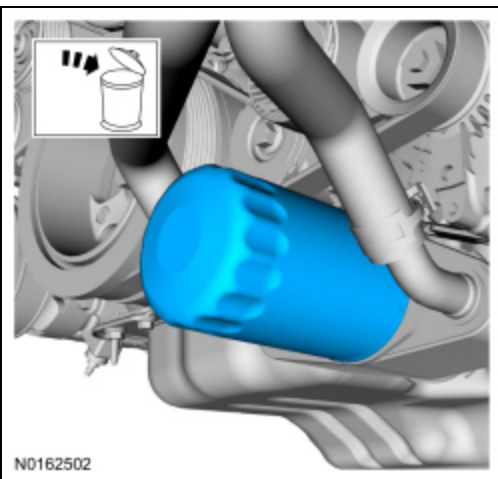
5. Connect the LH CMP sensor electrical connectors.



6. Install the **LH** exhaust manifold. Refer to Exhaust Manifold — RH.
7. Install the **LH** hydraulic lash adjusters. Refer to Hydraulic Lash Adjuster.
8. Install the intake manifold. Refer to Intake Manifold.
9. Install the **ACL**. Refer to Section 303-12.
10. Remove the oil pan drain plug and drain the engine oil.
 - Install the oil pan drain plug.
 - Tighten to 26 Nm (19 lb-ft).



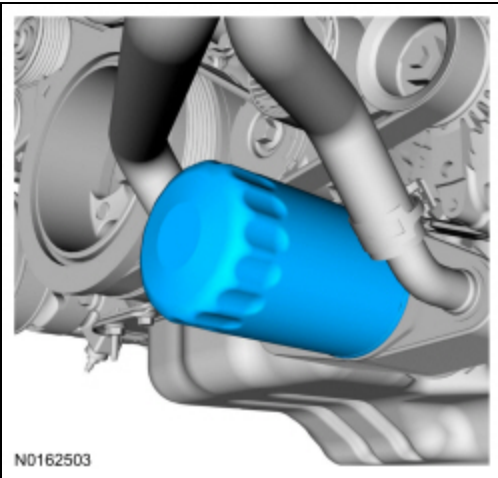
11. Remove and discard the oil filter.



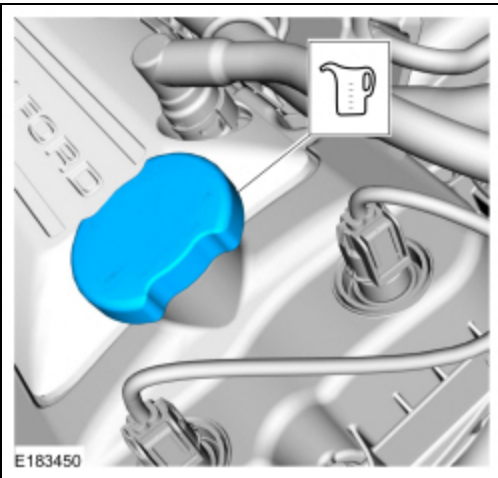
12. **NOTE:** Do not lubricate the O-ring seal.

Install a new oil filter.

- Tighten in 2 stages:
 - 5 Nm (44 lb-in).
 - additional 180 degrees.



13. Fill the engine with clean engine oil.



14. Fill and bleed the engine cooling system. Refer to Section 303-03.

15. After completing the repairs, perform the Misfire Monitor Neutral Profile Correction procedure using a diagnostic scan tool.